

Robot Operation Manual

This manual is applicable several types of robot, such as RH robot, RA light load robot and RA medium load robot, etc. And for other special robots, please refer to other relevant instructions.



Please ensure that the relevant instructions reach the end user of this product.

CROBOTP Relevant instructions:

CRP Robot Safety Manual

Programming Instruction Manual

CRP-G4-CD60 Electric Cabinet Manual

CRP-G4-CD80 Electric Cabinet Manual

RH Robot Mechanical Manual

RC Robot Instruction Manual

RA-L Robot Mechanical Manual

RA-M Robot Mechanical Manual

RH Robot Maintenance Manual

Thank you very much for choosing our products!

Please keep the relevant manuals of this product properly for reference when necessary !

If the equipment needs to be transferred, please hand over the relevant information to the other party !

Keys, functions and options not described in the robot related manual are deemed to be unavailable and should not be used !

Revision description:	
2021-12-17	first draft
2022-10-25	Modify cover

Description of safety warning signs

Please read carefully and understand totally of these instructions and other related documents before using this system. Please start using the system only after mastering all equipment knowledge, safety knowledge, and notices.

Safety attentions in this instruction are graded into four categories- DANGER, WARNING, BEWARE, and Caution.

Sign	Description
 DANGER	It means that if the logo is ignored and the robot is used improperly, death or serious injury may be caused.
 WARNING	There is danger during misoperation, which may cause moderate injury or minor injury accidents and equipment failure.
 BEWARE	Failure to observe the contents of this sign may cause personal injury and / or mechanical damage.
★ Caution	Indicates attention information about robot specifications, operation and maintenance, etc.

It should be noticed that things labled with "Caution" may also result into severe consequences in different situation. So it' s very important to pay attention to every "Caution" instructions, and stay strictly with them.

Some instructions even may not be labled with "DANGER" or "Caution", but users should also obey them.

Safety Attentions

! DANGER

★ Before operating the robot, press the emergency stop button on the teaching pendant and make sure main power supply of servo is off and motor is in "off-power and brake" status. After cutting off the servo power, servo power indicator on the teach pendant is red.

Under an emergency, personal injury or equipment damage may result if robot cannot be stopped in time.

Emergency stop button



★ If you want to remove the emergency stop, you should first remove the accident causing the emergency stop, and then connect the servo power supply.

Personal injury may result due to the unwanted robot movement from wrong operation.

rotate the button according the white arrow to remove the emergency stop.



★ When teaching within the robot movement region, please obey the following principle:

Keep observe the robot from the front.

Follow the operation manual strictly.

Consider the backup plan if robot move suddenly toward yourself.

Set up the shelter just in case.

Personal injury may result due to the unwanted robot movement from wrong operation.

★ When following processes are operated, make sure nobody is in the robot movement region and operator is in the safe position:

The robot control cabinet is powered on.

Controlling the robot with the teaching programmer robot

Test run of robot

Reoccurrence automatically

Personal injury may result when entering the robot movement region or contacting the robot.

Press the emergency stop button immediately when exception occurs.

CAUTION

★Following items need confirming before operating robot:

Operator is well-trained with robot-operation;

Operator has adequate knowledge of the movement feature of robot;

Operator is fully aware of the danger of robot;

Operator hasn't drink alcohol;

Operator hasn't take drugs that affect the nerve system and make reaction slow.

★Following items need checking before robot teaching process.
If any abnormal is found, fix it or take other necessary measurement immediately.

Movement of robot is normal or not;

Zero point adjustment is correct or not;

Auxiliary equipment connected to robot is normal or not.

★Please put the teach equipment to original position after usage, and place it firmly.

Please don't put the teach programmer on robots, claps or ground. Otherwise, personal injury or equipment damage may result when teach programmer has collision with robot or claps.

Please be cautious with the incidental falling of teach programmer, which may leads to wrong movement of robot, personal injury and equipment damage.

 MUST

★All the operators of robot systems should participate in system training, study safety protection measurements and understand robot functionality.

★Make sure no abnormal or dangerous situation of robot and auxiliary equipment occurs, before starting operating robot.

★ Unplug the power or press emergency stop button before entering the operation region, even robots aren' t running.

★ When programming within the movement region of robot, corresponding guard should be set to make sure that robot can be stopped under emergency circumstances. Please don' t wear gloves when teach and inch the robot. Please take the slow mode when inching the robot, in case emergency occurs and robot can be stopped efficiently.

★Operators must be familiar with the emergency stop buttons on the robot controller and periphery controller, to the degree that these buttons can be pressed exactly under emergency.

★Never think that static status of robot means the finish of program. Static robot is likely to receive the input signal to get moving.

Icon description

The keys or icons to be used in the text will be represented by the marks in the following table. The physical keys are represented by "<>" and the screen keys are represented by "【】".

Marks	icons	Description
< ⏻ >		Run button
< II >		Pause button
< ■ >		Interface switching button
【R】		Reset key
【▼】 \ 【▲】		Click to open hidden items
【Servo down】		Servo power down display status, click to switch to servo power on status.
【Servo on】		Servo power on display status, click to switch to servo power down status.
【Axis prohibition】		Prohibit axis movement
【Key movement】		Move each axis by pressing the key.
【Expansion axis】		Click to control the expansion axis
【Joint coordinates】		Click to use joint coordinates
【Tool coordinate】		Click to use the tool coordinate system
【User coordinate】		Click to use user coordinate system
【Rectangular coordinates】		Click to Rectangular coordinates system
【Single line running】		Click to enter single line running mode
【Single-cycle running】		Click to enter Single-cycle running mode
【Infinite loop mode】		Click to enter infinite loop mode
【Auto speed 10%】		The auto speed of the robot is 10%
"Light yellow"		
"Cyan"		

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1. Product Description

Overview of industrial robots

Components of industrial robots	
1	Robot
2	Electric control cabinet
3	Teaching Pendant
4	Connecting cable
5	Software
6	Other auxiliary equipment

Composition of industrial robots (Fig 1.1):

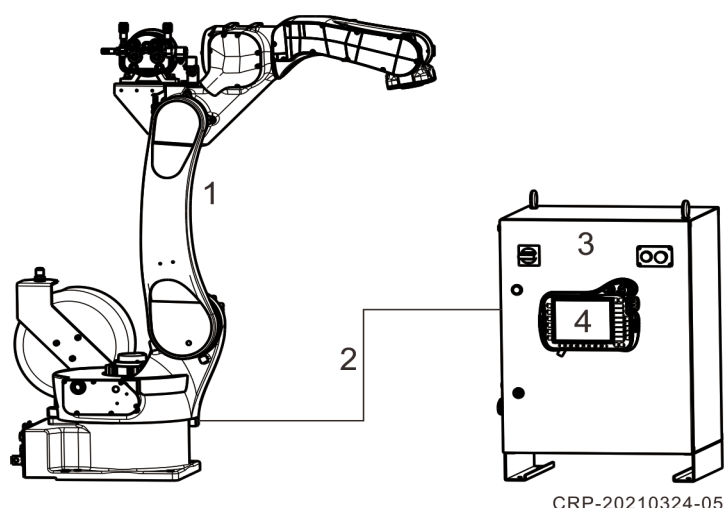


Fig.1.1

1-Robot

2-Connecting cable

3-Electric control cabinet

4-Teaching pendant

2. Safety

2.1 Responsibility statement

Industrial robots meet the current technical level and current safety technical regulations. Illegal operation of industrial robots will cause personal injury and damage to robots and peripheral equipment.

The industrial robot shall not be modified without the consent of CRP . Other auxiliary equipment not within the supply scope of CRP company may also be included in the industrial robot. If the auxiliary equipment causes damage to the industrial robot, the responsibility shall be borne by yourself.

2.2 Use safely

For safe use of the robot, please read CRP Robot Safety Manual, CRP User Manual and other relevant documents. Otherwise, serious accidents may be caused.

Relevant operators of industrial robots must receive relevant professional training, and those who fail to pass the training shall not take up their posts.

he working area must be set before the industrial robot is used, and the working area must be set with safety fence, safety lock, safety grating and other safety measures (Fig. 2.1).

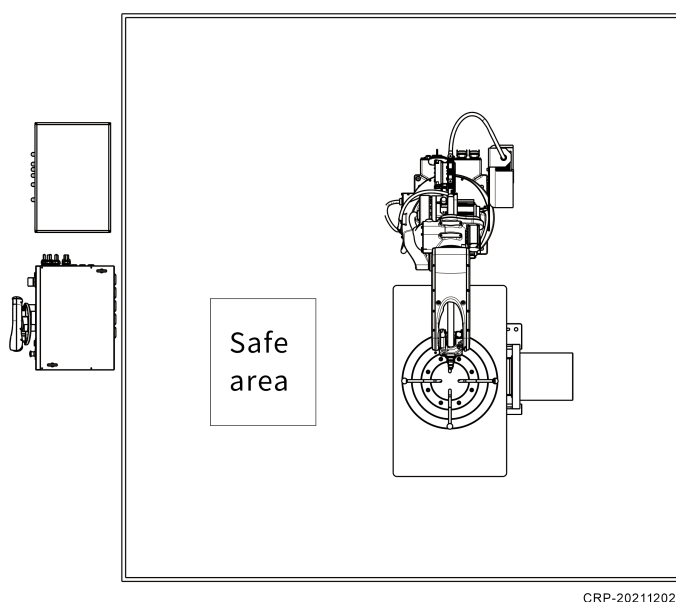


Fig.2.1

3. Installation

For the installation of industrial robots, please read the mechanical instructions of robots and other relevant documents in detail. Otherwise, serious accidents may be caused.

3.1 Installation precautions

During the test, please hang a warning sign of "Equipment test, No entry" outside the safety door.



3.2 Installation robot

1. Installation of electric control cabinet:

When installing the control cabinet, keep away from the heat source and the wall to make the air smooth without affecting the heat dissipation. It shall be installed outside the safety fence without affecting the opening and closing of the cabinet door. The robot body can be clearly seen at the installation position of the cabinet door, but it should not be too far away from the robot body.

2. Robot body installation:

Due to the large weight and fast movement speed of the robot body, the installation position of the robot needs to bear a large load. A concrete foundation of more than 200mm is required for ground installation. If the foundation thickness is less than 200mm, steel plates shall be added and fixed with anchor bolts. When the foundation is uneven, the four corners shall be padded with gaskets with a thickness of more than 100mm. The installation shall not be too far away from the control cabinet.

3.3 Cable connection

CRP robot comes with its own connecting cable when it leaves the factory. Read relevant manuals carefully and connect wires as required.

In order to prevent electric shock accidents, the robot body, electric control cabinet and peripheral electrical equipment shall be reliably grounded separately before power on.

Ensure the wiring safety of the power line with high voltage, the equipment wiring shall be neat and orderly, the cable and wire routing shall be in the wire groove, and the cable length shall be reasonably selected according to the different functions of the equipment and cables.

The robot control cabinet of CRP is equipped with safety board. Please refer to the instructions of relevant robot electric cabinet for safe wiring.

It is forbidden to plug and unplug the connector with power.

4. Teaching Pendant

CRP-TPHK80K-C teaching pendant is a hand-held programmer used to control industrial robot of CRP. The teaching pendant has various operation and display functions required for the operation and programming of CRP industrial robot.

CRP-TPHK80K-C teaching pendant is equipped with a touch screen (resistance screen), it can be operated directly with fingernails and its own touch pen. Generally, no mouse or keyboard is required.

4.1 Introduction to teaching pendant



Fig. 4.1

- 1-Alarm indication: If the robot gives an alarm, the red light is on.
- 2-Power on status: the green light is on when the teaching pendant is powered on
- 3-Display / touch screen: displays the status of the robot
- 4-Mode switches:TEACH, PLAY, REMOTE
- 5-Emergency stop button: Press the button to stop the robot.
- 6-Handwheel: control cursor, in menu list, parameter interface, variable table, etc
- 7-Teaching pendant cable: connect the teaching pendant to the robot electric control cabinet



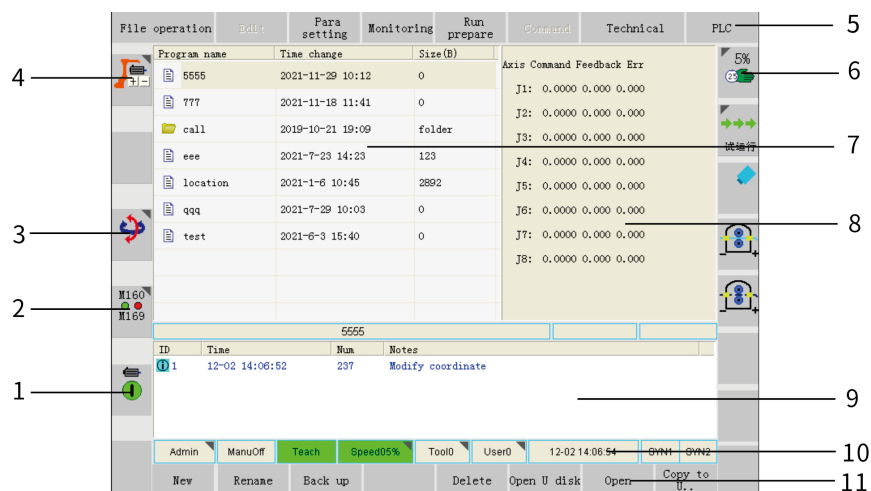
Fig.4.2

- 1-Touch pen: operate the touch screen of the teaching pendan
- 2-Safety switch (holding switch): control the motor holding brake when moving
- 3-USB (reserved)

4.2 Teaching pendant interface

The display part of the teaching pendant is an 8-inch color display screen plus a touch screen. It is used to display the robot operation interface and carry out corresponding operations.

The display interface is mainly composed of three display areas (general display area, monitoring area and information prompt area), and the main menu, coordinate area, status display and sub menu are distributed around, as shown in the following figure.



CRP-20211202-02

Fig.4.3

1-Servo on / off

2-M shortcut key window on / off

3-Robot coordinate system selection

4-Robot movement way selection

5-Main menu area

6-Robot speed regulation

7-Universal display window

8-Monitor window

9-Message prompt window

10-Status window

11-Sub menu bar

The three display areas can be switched by pressing the < ■ > key or directly clicking the screen to switch and activate the display area.

When a display area is selected, the background of the area will change or the cursor bar will appear. When the display area switches, the status control, coordinate area and sub menu will change.

Activation status of general display area: Blue cursor bar is displayed when the program list is displayed; When the program is open, the background is "cyan".

Monitoring area activation status: the background is "cyan".

Active status of information prompt area: Blue cursor bar is displayed.

Among the three display areas, the monitoring area can be closed. When the monitoring area is displayed, the general display area will be automatically reduced to half width display; After the monitoring area is closed, the general display area will be automatically enlarged to the whole display.

The main menu can only be operated by clicking on the screen.

The status control area, coordinate area and submenu area can be switched through the corresponding buttons on the outside of the screen or directly clicked on the screen; When clicking, the button with corner mark will pop up a window, and there is no corner mark button to switch the state

The icon or area with triangular corner mark 【▴】 \ 【▾】 can be clicked and a dialog box will pop up.

When the safety switch is pressed, the "light yellow" button (mobile key of robot) in the coordinate area and sub menu area can move the robot.

Description
CRP-TPHK80K-C teaching pendant can be operated by keyboard and touch. The status display area, coordinate area and sub menu area can be operated by physical keys. At the same time, all buttons, input boxes, options, etc. can be clicked.
The physical key operation and click operation in the status display area, coordinate area and sub menu area are different; Click to switch status and pop up dialog box; The physical key will not pop up a dialog box, but only change the state.
After clicking the input box, the system will automatically recognize numbers or characters and pop up the corresponding full virtual keyboard or numeric keyboard.
The status next to the keys of all robots that can act is displayed in light yellow. At the same time, only physical keys can act. Clicking is invalid.

4.3 Use of teaching pendant

1. Place your left arm between the teaching pendant cable and the armrest, hold the side armrest of the safety switch of the teaching pendant with your palm, and place your index finger and middle finger on the safety switch. (as shown in Fig. 4.4).



Fig.4.4

2. Lift the teaching pendant with your left hand, turn it over, the display interface is upward, and hold the teaching pendant in a suitable position on your abdomen. Operate the touch screen, keys, switches, etc. of the teaching pendant with the right hand. (as shown in Fig.4.5)



Fig.4.5

3. Standing position: the operator shall stand outside the working area of the robot, and the robot body shall be within the field of vision of the operator. In case of emergency, press the emergency button immediately to stop the robot action.

5. Robot Zero Point and Coordinate System

For operation of industrial robots, please read the CRP Robot Safety Manual , CRP User Manual and other relevant documents in detail. Otherwise, serious accidents may be caused.

5.1 Robot Zero Position

Only after the zero point is correctly calibrated, the industrial robot can achieve the best point accuracy and Trajectory Accuracy, and fully adapt to the programming operation.

CRP industrial robots go through strict zero check before leaving the factory. Please check carefully before using. The zero position attitude is shown on the left of Fig.5.1, and the zero label style is shown on the right of Fig.5.1..

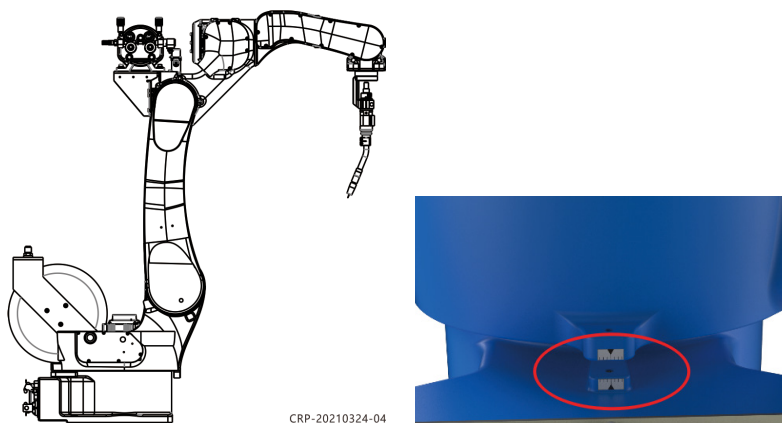


Fig.5.1

5.2 Coordinate system

Joint coordinates: the robot moves independently based on each axis.

Cartesian coordinate: the robot moves based on the default Cartesian coordinates XYZABC.

Tool coordinates: the robot moves based on the calibrated tool coordinates XYZABC.

User coordinates: the robot moves based on user-defined user coordinates XYZABC.

5.2.1 Tool coordinate

Tool coordinate system can be established according to the use of industrial robots, and several tool coordinate systems can be established.

Six-point calibration: The robot records data individually close to the target point through six different postures, and the robot can automatically calculate the position of tool control point and save it in corresponding tool files.

Use a tool to verify that the input is the coordinate value of the tool control point in the flange coordinate. (Fig. 5.2)

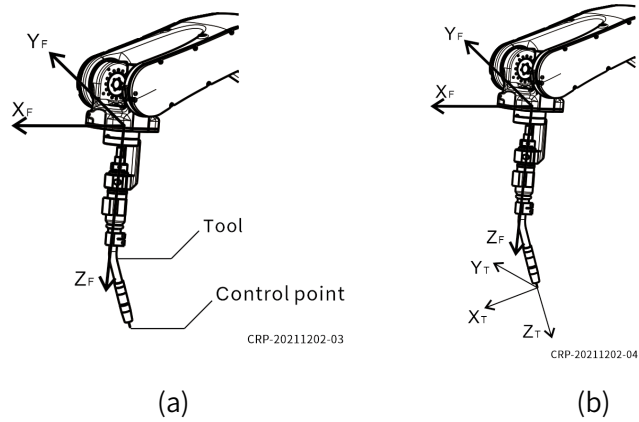


Fig.5.2

(a) Tool dimensions are based on robot end coordinates

(b) Relationship between tool coordinates and end flange coordinates

The six point checkpoint method (Fig. 5.3).

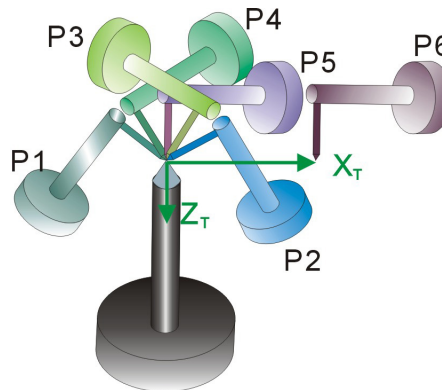


Fig.5.3

The steps for setting the tool coordinate system are as follows:

Click **【Run prepare】** - **【Tool coordinate】** (Fig.5.4)

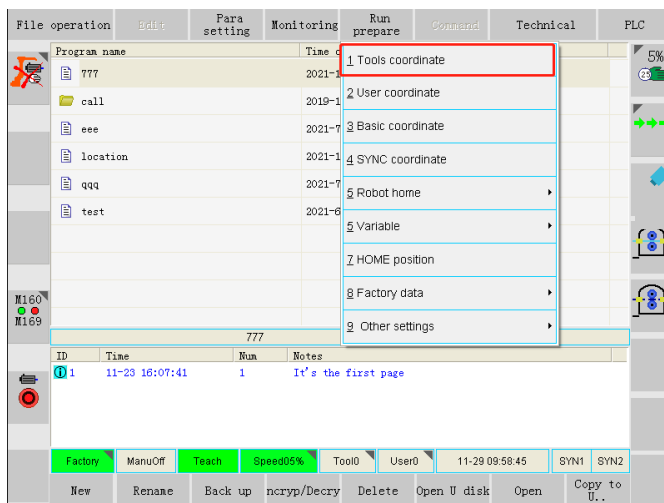


Fig.5.4

Enter the interface to set the tool coordinate system. The tools coordinate system setting interface is shown in the following figure.

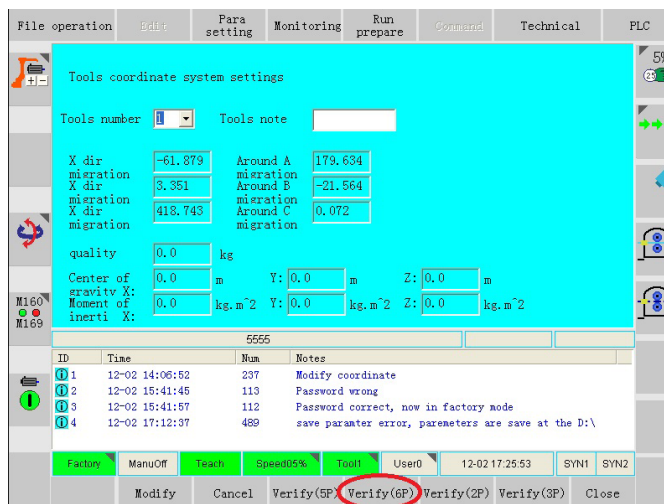


Fig.5.5

As shown in Fig. 5.5, select the tools number and click 【Verify(6P)】 to enter the tool coordinate system calibration interface as shown in Fig. 5.6.

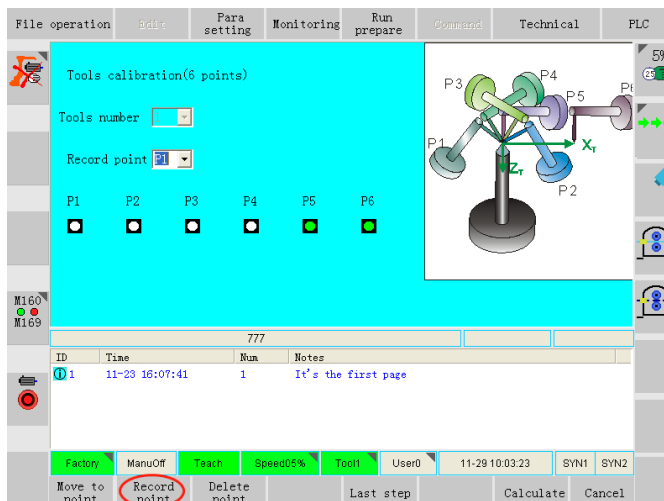


Fig.5.6

In the interface shown in Fig.5.6, select the recording point number in sequence, move the TCP cusp to the corresponding position, and click 【Record point】 , and the corresponding point number indicator will turn green.

Description
As shown in Fig.5.7, P1-P4 points approach the TCP cusp in different attitudes. The greater the attitude change, the more conducive to TCP calculation. When checking P5 point, the TCP cusp must be perpendicular to the gun calibrator. And P6 point is used to determine the X direction of tool coordinates, that is, the connecting line between P5 point and P6 is the X direction of tool coordinates.

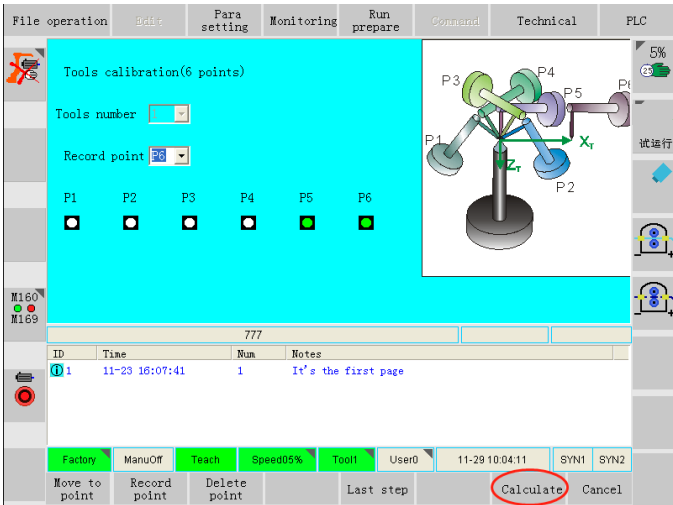


Fig.5.7

Click calculate in the interface shown in Figure 5.7. The system will automatically calculate the current tool coordinates and determine the tool coordinate system and direction. And press 【Cancel】 in the interface shown in Fig.5.7 to obtain the size of the tool tip relative to the flange coordinate at the end of the robot.

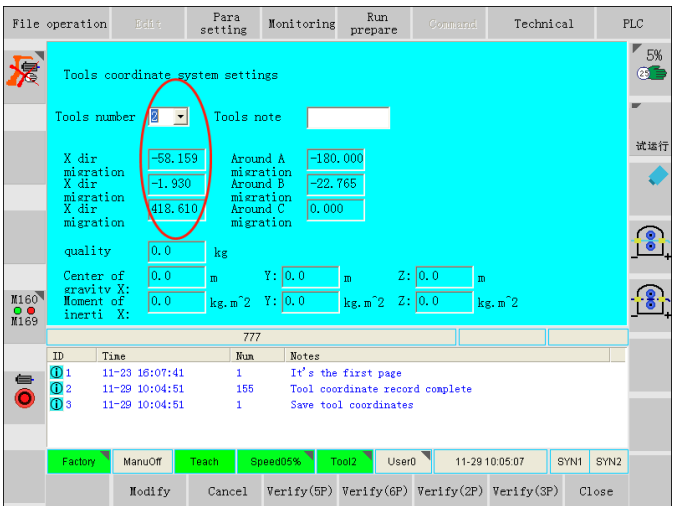
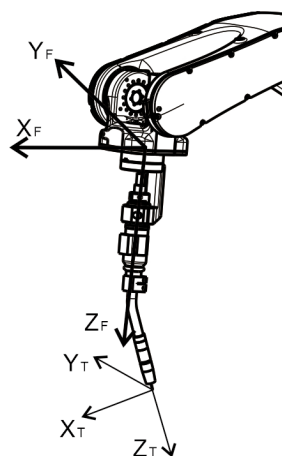


Fig.5.8

Description
The tool size is determined according to the coordinate system shown in the above figure (Fig.5.8).



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Fig.5.9 Relationship between tool coordinates and end flange coordinates

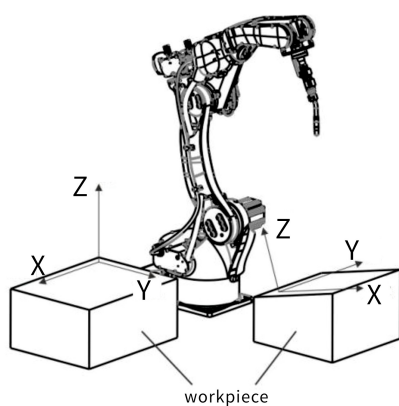
The tool coordinate system is shown in Fig.5.9, which is based on the Cartesian coordinates of the tool tip.

After calculating the tool coordinate system, you can switch to **【Tool coordinates】** under the tool coordinate system to verify the tool coordinate system and direction.

After verification, click **【Close】** to exit.

5.2.2 User coordinate

As shown in Fig.5.10, the user (workpiece) coordinate system can be established according to the use of industrial robots, and multiple user (workpiece) coordinate systems can be established.



CRP-20211203-01

Fig. 5.10

Click **【Run prepare】** - **【User coordinate】** , as shown in the following figure.

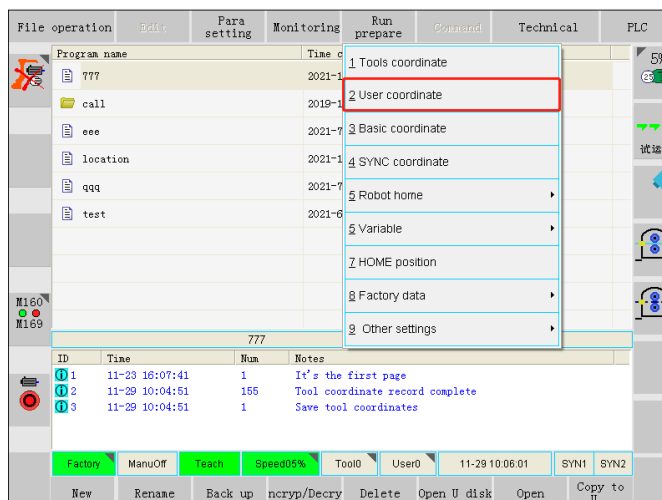


Fig.5.11

Enter the user coordinate system setting interface to set relevant parameters, as shown in the following figure.

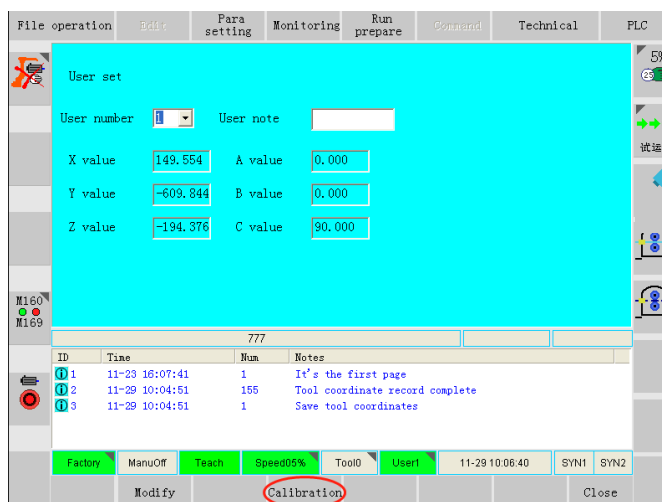


Fig.5.12

After selecting the user number, click **【Calibration】** to enter the user coordinate calibration interface, as shown in Fig.5.13.

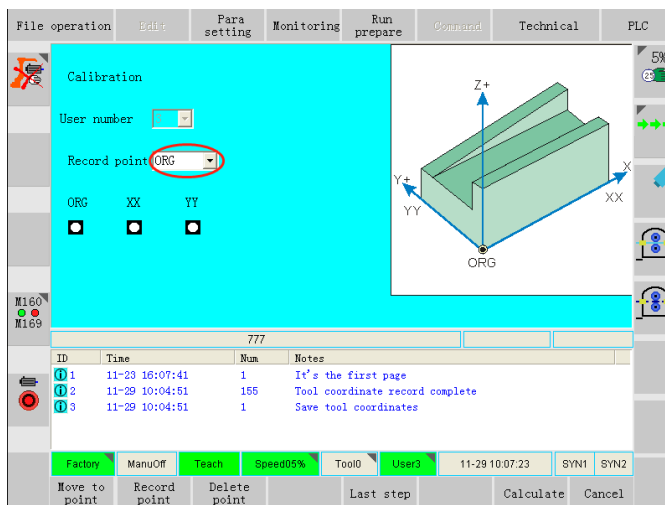


Fig.5.13

First, set the origin "ORG" of the user (workpiece) coordinate system, and move the robot tip (such as the welding wire on the welding gun, the positioning pin on the fixture, etc.) to the workpiece origin. Then click **【Record point】** to record the origin of user (workpiece) coordinates.

Select "XX direction" to determine the X axis, as shown in Fig.5.14.

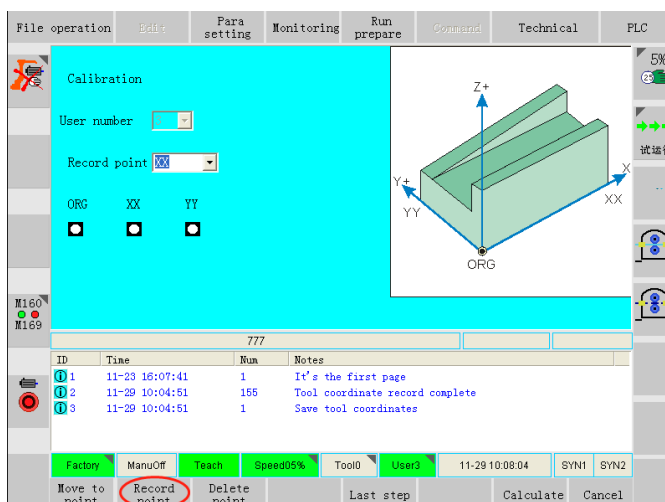


Fig.5.14

As shown in Figure 5.14, set the X direction of the user (workpiece) coordinate system, and move the robot tip along the X direction on one side of the workpiece for a certain distance. Then click the **【Record point】** button to record the XX direction of user (workpiece) coordinates.

Select "YY direction" to determine the Y axis, as shown in Figure 5.15.

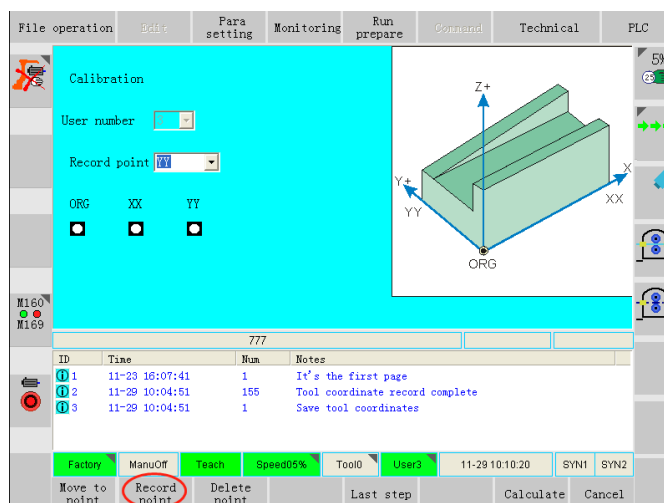


Fig.5.15

As shown in Figure 5.15, set the Y direction of the user (workpiece) coordinate system, and move the robot tip to the other side of the workpiece in the Y direction for a certain distance. Then click **【Record point】** to record the Y direction of user (workpiece) coordinates.

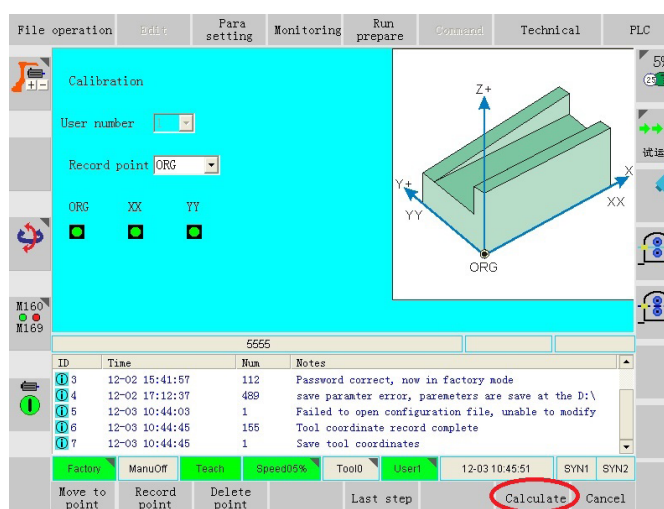
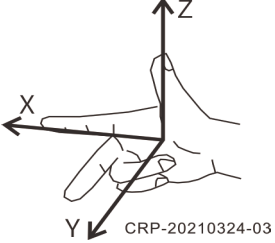


Fig.5.16

After determining the origin, XX direction and YY direction, click **【Calculate】** in the interface shown in Figure 5.16, and the system will automatically calculate the coordinates of the current user (workpiece) and determine the coordinate system and direction on the workpiece.

Description
the establishment of user coordinate system refers to the right-hand rule (as shown in Figure 5.17), and the positive direction of Z is in the thumb direction of X-Y rotation. When establishing workpiece coordinates, the positive direction of Z is usually far away from the workpiece. Therefore, when establishing workpiece coordinates, it is necessary to consider which edge in X and Y directions is respectively.  Fig.5.17

After the user coordinate system is calculated, click **【User coordinate】** to switch to the user coordinate system to verify whether it is the desired user coordinate direction. After verification, click **【Cancel】** to exit.

6. Manual Operation

Before operating industrial robots, please read the CRP Robot Safety Manual, CRP User Manual and other relevant documents in detail. Otherwise, serious accidents may be caused.

6.1 Pre operation inspection

Before using the equipment, check whether the equipment has other personnel use information, whether the robot has alarm, confirm that there are no other irrelevant personnel in the working area of the equipment, and the emergency stop button, safety switch, safety lock, safety grating, sensor, etc. are effective.

6.2 Single axis operation

1. Turn on the host.

At this time, the robot control system is turned on. (as shown in Fig.6.1):

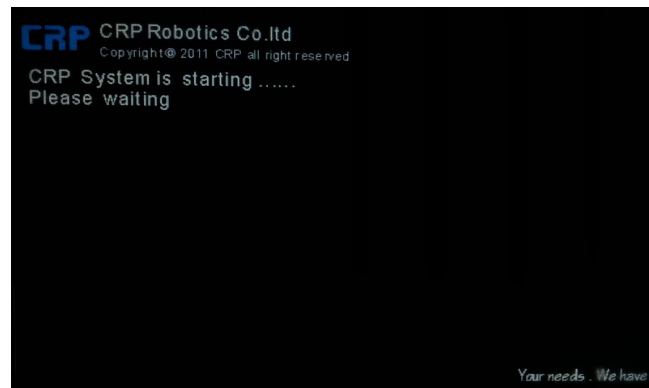


Fig.6.1

Wait for the system to start up, as shown in Fig.6.2. If there is an alarm, please click the **【R】** to reset the alarm. If the alarm cannot be reset, please check the corresponding circuit or other settings according to the information prompt.

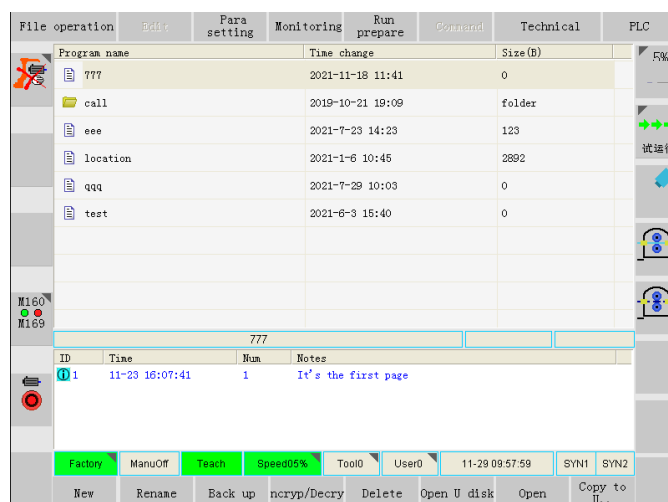


Fig.6.2

2. Turn the mode switch to "TEACH", as shown in note 3 in Fig.6.3.

3. Click the corresponding icon or icon corresponding button.

Click **【Servo down】** to change it to **【Servo on】** ; Click **【Axis prohibition】** to change it to **【Key movement】** ; Click **【Cartesian coordinates】** / **【User coordinates】** / **【Tool coordinates】** to change it to **【Joint coordinates】** .

4. After the above operations are completed, Press the teaching pendant safety switch and keep the second gear closed, the teaching pendant will display as shown in the figure below.



Fig.6.3

1-Robot axis coordinate display 2-Robot axis movement key

Press the physical key (axis key) on the left side of the corresponding axis, and the robot will make corresponding actions corresponding to the single axis. Carefully observe the movement direction of the robot to prevent safety accidents.

6.3 Run in coordinate system

1. The first 4 steps of the operation steps of the moving robot in the coordinate system are consistent with the single axis motion. See "6.2 single axis operation 1-4 steps".

2. Click the corresponding icon or icon corresponding button

Click **【Servo down】** to change it to **【Servo on】** ; Click **【Axis prohibition】** to change it to **【key movement】** ; Click **【Cartesian coordinates】** / **【User coordinates】** / **【Tool coordinates】** to change it to **【Joint coordinates】** .

3. After the above operations are completed, press the safety switch of the teaching pendant and keep the second gear closed, and the teaching pendant will (as shown in Fig. 6.4) display as follows.



Fig.6.4

1-Robot axis coordinate display 2-Robot axis movement key

Press the physical key (axis key) on the left side of the corresponding axis, and the robot will make corresponding actions corresponding to the single axis. Carefully observe the movement direction of the robot to prevent safety accidents.

7. Program editing

For industrial robot programming, please read CRP Programming Instruction Manual and other relevant documents. Otherwise, serious accidents may be caused.

7.1 New-built program file

Edit steps:

1. Turn the mode switch to "TEACH".
2. Select the appropriate tool coordinate system.
3. Enter the program list interface. (Figure 7.1)

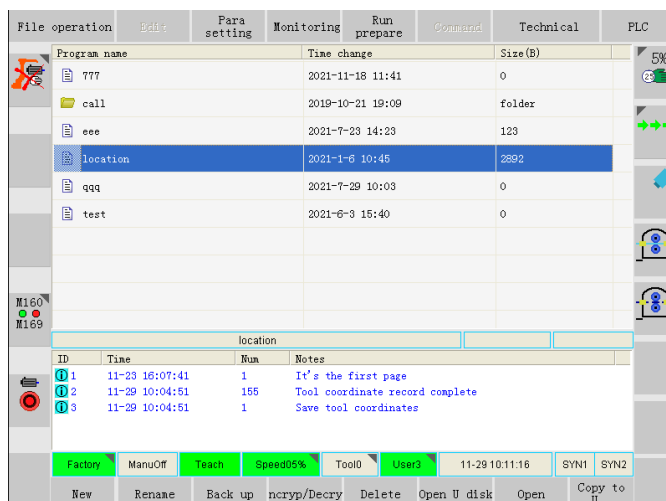


Fig.7.1

4. Click 【New】 in the submenu (Fig.7.1).

5. Enter the name of the new program in the pop-up window (e.g. 5555). After entering the file name, as shown in Fig.7.2.

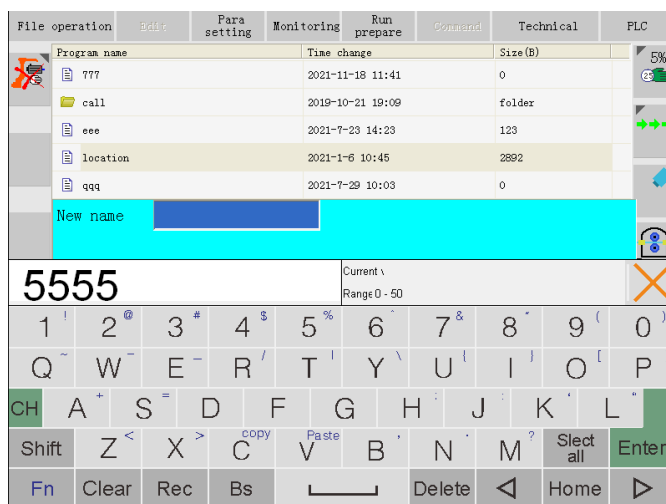


Fig.7.2

Click twice to confirm and the newly created program "5555" will be displayed in the program list interface, as shown in Fig.7.3.

000	2021-8-19 09:25	58
1234	2021-9-3 11:24	0
5555	2021-9-3 11:29	0
c1111	2021-9-3 11:25	0

Fig.7.3

7.2 Programming

1. Click the **【Open】** icon in the submenu area to open the "5555" program and enter the program editing interface, as shown in Fig.7.4.

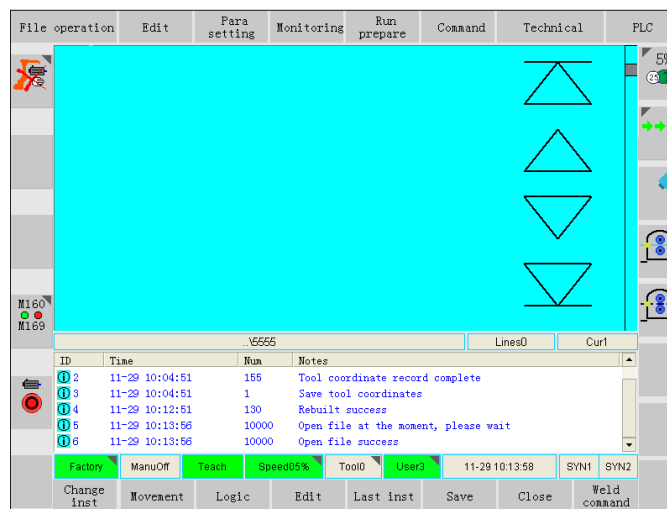


Fig.7.4

Program cursor movement key Description:

⏮ Move the cursor to the first line of the screen;

⬆ move the cursor up one line;

⬇ move the cursor down one line;

⏭ Move the cursor to the last line of the screen;

2. Through the axis key on the teaching pendant, move the end of the robot to the position of program point 1. Click the sub menu **【Movement】** icon to pop up the following window, and click the **【MOVJ】** icon.

Or click **【Commad】** - **【1 Movement】** - **【1 MOVJ】** in the main menu, as shown in Fig.7.5.

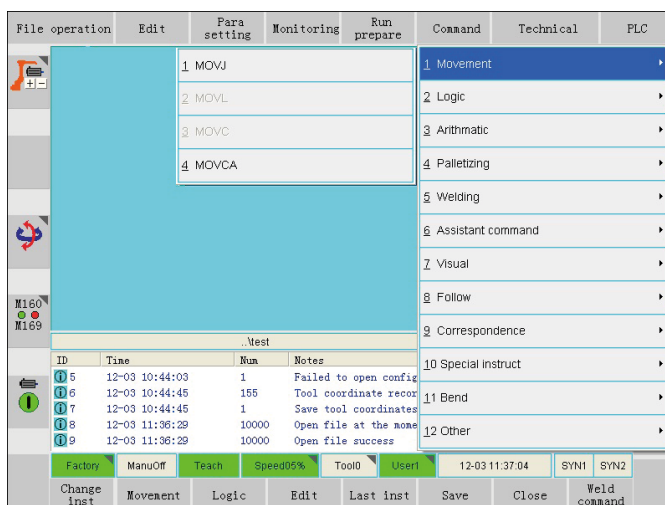


Fig.7.5

The command editing window pops up, as shown in Fig.7.6.

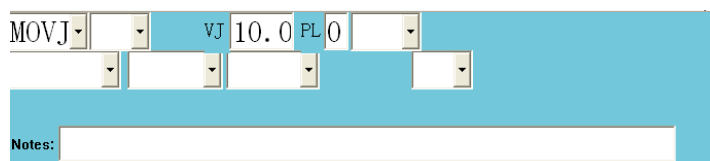


Fig.7.6

You can edit comments directly in the blank box after "Note".

If you need to modify the data, directly click the data area, and the numeric keypad will pop up automatically, as shown in Fig.7.7. Click "Enter" after input.

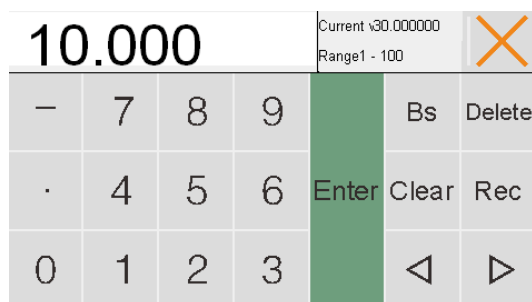


Fig.7.7

After inputting corresponding parameters as required, click 【Inst correct】. This command line is displayed in the program editing window. As shown in Fig.7.8.



Fig.7.8

The instruction editing of program point 1 is completed.

3. Move the end of the robot to the position of program point 2 through the axis key on the teaching pendant. Click the submenu **【Movement】** icon to pop up the following window, and click the **【MOVJ】** icon.

Or click **【Command】** - **【1 Movement】** - **【1 MOVJ】** in the main menu.

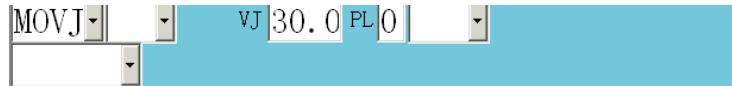


Fig.7.9

After inputting corresponding parameters in the window as required, click **【Inst correct】**, and the command line will be displayed in the program editing window, as shown in Fig.7.10.



Fig.7.10

★Caution	
Press the safety switch and the 【Inst correct】 key at the same time to record the point information. The same operation is required for point modification, otherwise only the command data or command can be modified.	

The instruction editing of program point 2 is completed.

4. Repeat step 2 or 3 to edit program point 3. Then change VJ speed to 25% and PL value to 0. Enter the command line of program point 3, as shown in Fig.7.11.

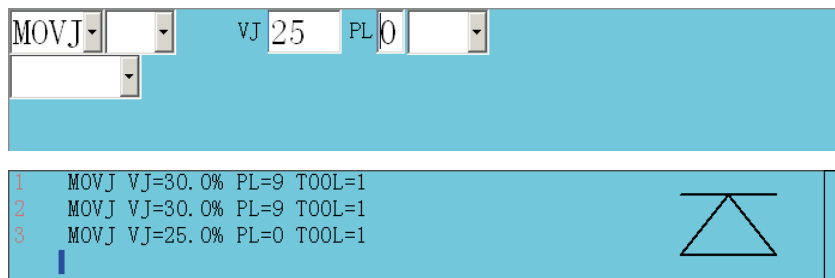


Fig.7.11

5. Click **【Commad】** - **【5 Weld】** - **【1 ARCSTART】** in the main menu to open the following window, as shown in Fig.7.12.



Fig.7.12

After inputting corresponding parameters as required, click the 【Inst correct】 key, and the command line will be displayed in the program editing window, as shown in Fig.7.13.

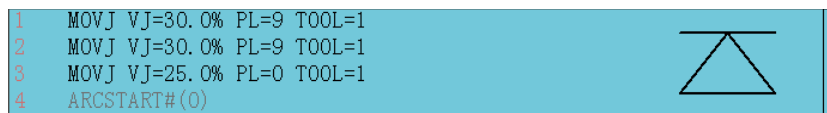


Fig.7.13

6. Repeat the above similar steps. Input all program points and instructions, as shown in Fig.7.14.

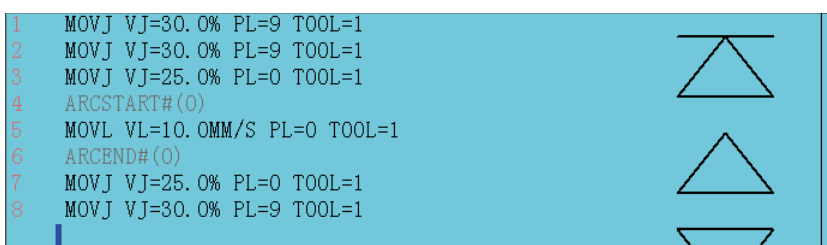


Fig.7.14

7. Click 【Save】 in the submenu area, and then click 【Close】 to close the program editing interface.

Through the above steps, the instance program is created.

★Caution

In the infinite loop running, if the running program has no motion instructions, only logic instructions. The "TIME" instruction needs to be added at the end of the program, and the delay value is set to 10ms.

In the infinite loop running, if the program has only one line of motion instruction, the "TIME" instruction needs to be added later, and the delay value is set to 10ms.

In the play mode, if the unconditional jump command is used at the end of the running program, and the robot has no action command or all motion commands. At the end of the program, the time instruction needs to be added, and the delay value is set to 10ms.

The last line of the program does not need to jump to the first line without unconditional jump instruction, and the running mode can be switched to infinite loop.

7.3 Modify program

Based on the edited program line, the program instruction needs to be modified. The specific operations are as follows:

Select the program line to be modified, and then click **【Change inst】**, as shown in Fig.7.15.

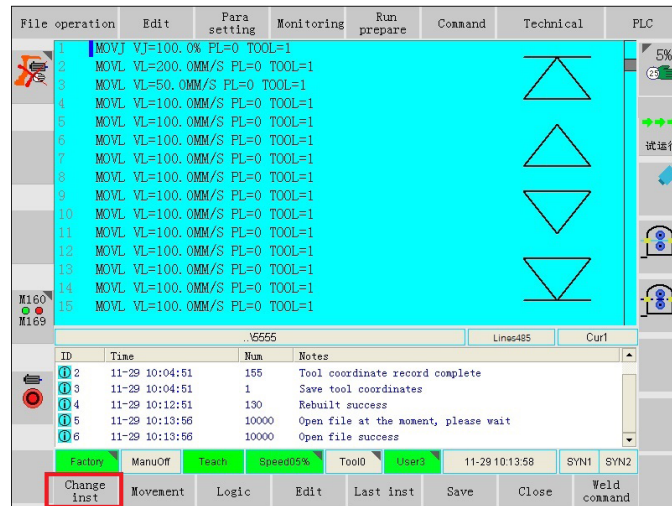


Fig.7.15

After adjusting the parameters, commands and points to be modified, press and hold the safety switch and click **【Inst correct】**, the program modification is completed, as shown in Fig.7.16.

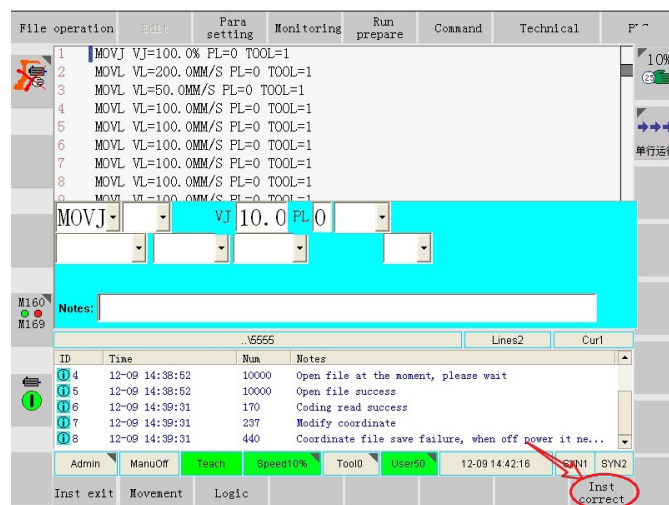


Fig.7.16

8. Program Running

8.1 Program commissioning

1. Switch to **【Key movement】** state and **【Servo on】** state.

2. Adjust the manual speed to an appropriate speed. It is recommended that the speed ratio after adjustment should not exceed 10%. The adjusted speed multiplier displays 10% in the status display area or status control bar, as shown in the figure below. The speed can be modified at the two positions circled on the figure.

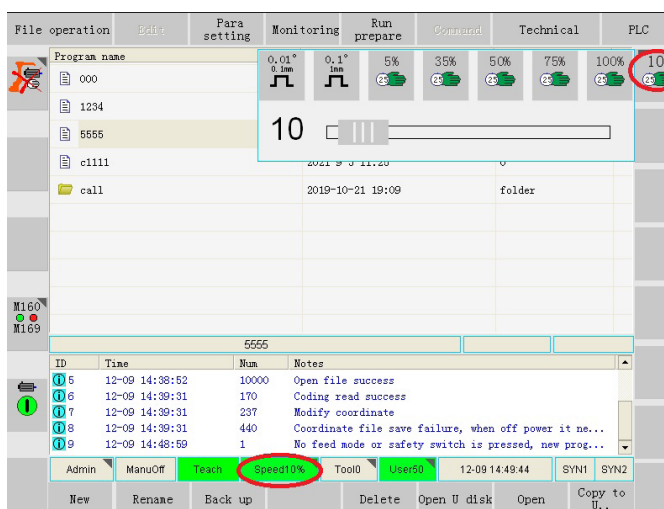


Fig.8.1

Step:

1. Select "5555" program, as shown in Figure 8.2.

Program name	Time change	Size(B)
5555	2021-11-29 10:12	0
777	2021-11-18 11:41	0
call	2019-10-21 19:09	folder
eee	2021-7-23 14:23	123
location	2021-1-6 10:45	2892

Fig.8.2

2. Double click the "5555" program or click the **【Open】** key in the submenu to open the program and enter the program editing interface. As shown in Fig.8.3.

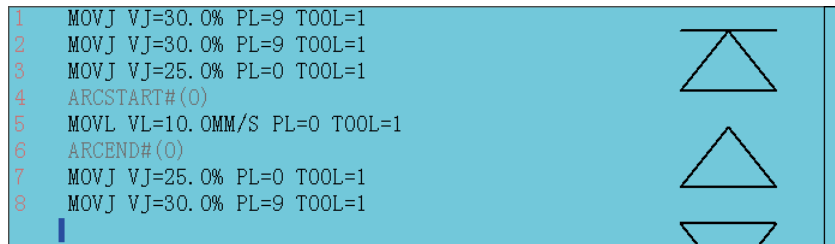


Fig.8.3

3. Click the cursor movement key in the screen to move the cursor to the program line to be tested, such as the second line, as shown in Fig. 8.4.

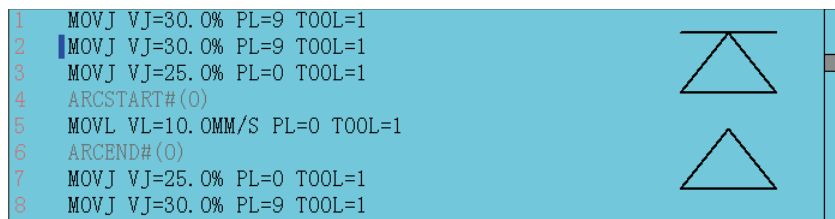


Fig.8.4

4. Press and hold the < Ⓢ > key continuously. The system controls the robot to execute the command of the line where the cursor is located. Such as robot action, such as robot action command, IO output command, operation command, logic command, etc.

★Caution

When the cursor is in the if, while and switch instruction structures, the system will prompt an error. Before commissioning, move the cursor outside the instruction structure.

8.2 Automatic running

If there is no error in the program commissioning of the program, the automatic operation of the program can be started. The status description is shown in the table below.

Icon	Code in the text	Description
	【Single line run】	Status display, currently single line running mode
	【Single loop】	Status display, currently single loop running mode
	【Infinite loop】	The status shows that it is currently running in an infinite cycle mode
	【Automatic speed 10%】	Status display, current automatic speed is 10%

★Caution

During the first automatic running, it is recommended to select the single line running mode, run line by line and observe the running. In the single line running mode, the rhythm will be slow, please pay attention!

When the single line running is correct, select the single cycle running mode. The running speed of the first single cycle program shall not exceed 5%, and gradually accelerate to the required process speed after multiple runnings are accurate.

After the single cycle running is correct, select infinite cycle running mode to start working.

The speed cannot be adjusted during program running. You need to press the stop key and then adjust the speed.

8.2.1 Single Line Running

► Switch the control mode switch to "PLAY", as shown in Fig.8.5.



Fig.8.5

The status bar displays "Play", then change the operation mode 【Single loop】 / 【Infinite loop】 to 【Single line run】 , and change the automatic speed to 【Automatic speed 10%】 .

After the previous preparations are completed, press < ◎ > to run the program, and a prompt box will pop up:

In the reproduction mode, press the Run button, you need to run the program, you need to press the Run button again, or click on the interface to give up.

Press < ◎ > again to run, and the operation interface is shown in Figure 8.6.

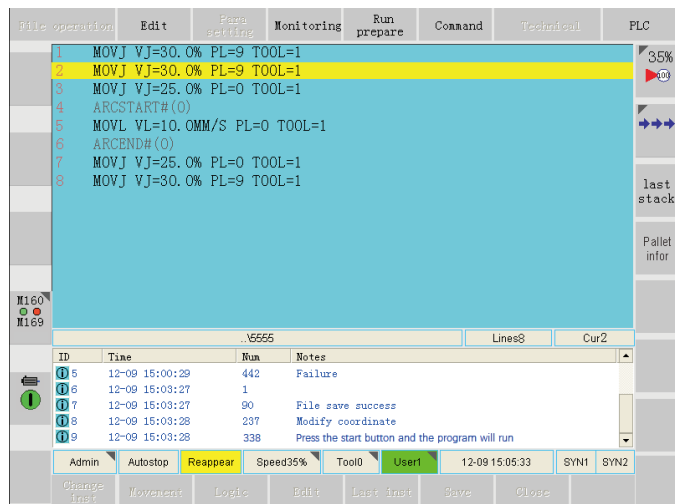


Fig.8.6

► Single step running :

When the robot runs the selected command line, the robot stops. To run the next command line, press < Ⓢ > again to run the program.

► Stop the robot during running:

If you need to pause (stop) during running, press the < || > key to slow down the system and stop program operation and robot action.

► Adjust speed:

During running, if it is necessary to increase the automatic speed, stop the robot first, and then operate the speed regulation button to realize the speed regulation function.

► Switching running mode:

1. During program running, you can directly click the running mode icon to switch back and forth between 【Infinite loop】 and 【Single loop】. When you need to switch to 【Single line run】, you need to stop the program before switching.

2. In the pause (stop) state, click the running mode icon, click "Select" in the pop-up window, or press the physical buttons < + > and < - > on the right corresponding to the mode icon to switch the running mode arbitrarily.

► When the program is running, the working mode is switched:

Currently in the play mode, if the program is running, you need to press the < || > key to stop the program. Then switch the mode switch to the desired mode (teaching mode or remote mode).

► Stop the program in play mode

In this mode, the program stops by pressing the stop key. The system slowly stop program running and robot action. After stopping the program in this mode, all internal states, output ports, counters, variables, etc. related to the program will be maintained. When starting again, directly press the < ◎ > key, and the program continues to execute normally.

★Caution

1. When the program running mode is "Single line run", after the program runs one line, the system slowly stops the program and robot operation. The system is in a static rather than stopped state. It is necessary to press the < I I > key to stop the program.
2. Switch the mode switch to TEACH mode or PLAY mode, and the program stops forcibly. During system processing, the pulse will be cut off, the enable will be closed and the holding brake will be opened directly. This method will cause robot impact and is not recommended.
3. If you need to pause (stop) during program running, press the < I I > key to slow down the system and stop program running and robot action. After stopping the program in this mode, all internal states, output ports, counters, variables, etc. related to the program will be maintained. When starting again, directly press the < ◎ > key, and the program continues to execute normally.
4. In order to determine whether the program stops, press the < I I > key and observe whether there is "Automatic stop" in the status bar .

8.2.2 Single Loop Run

► The toggle control mode switch is in PLAY mode , as shown in Fig.8.7.



Fig.8.7

The status bar shows "Play", and then changes the running mode **【Single line run】 / 【Infinite loop】** to **【Single loop】** , and changes the automatic speed to **【Automatic speed 10%】**

After the previous preparation is completed, press the < ◎ > key to run the program, and the system pops up a prompt box:

In the reproduction mode, press the Run button, you need to run the program, you need to press the Run button again, or click on the interface to give up.

Press < Ⓞ > again to run, the running interface is shown in Figure 7.8.

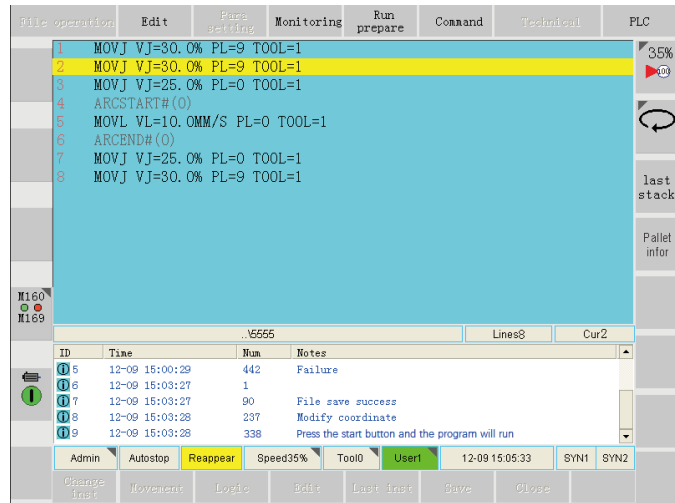


Fig.8.8

► Stop the robot during running:

During running, if you need to pause (stop), press the <II> key to slow down the system and stop the program and the robot action.

► Adjust speed:

During operation, if you need to increase the automatic speed, you need to stop the robot first, and then operate the speed control button to achieve speed control function.

► Switch mode:

1. While the program is running, you can switch back and forth between 【Infinite Loop】 and 【Single Loop】 by clicking on the running mode icon directly. When you need to switch to 【Single line run】 , you need to stop the program before you switch.

2. In the pause (stop) state, click the running mode icon, click "Select" in the pop-up window, or press the physical buttons < + > and < - > on the right side of the mode icon to switch the running mode at will.

► When the program is running, the working mode is switched:

Currently in Play mode, if the program is running, you need to press the <II> key to stop the program running, and then switch the mode to the desired mode (Teach mode or Remote mode).

► Stop program running in play mode

In this mode, the program stops running by pressing the stop key. System slowly stop program operation and robot action. When the program is stopped in this way, all internal state, output port, counter, variable, etc. related to the program are maintained. On restart, press the <◎> key directly, and the program will continue to run normally.

★Caution

1. Switch the mode switch to the Teach mode or the Play mode, and the program stops forcibly. When the system is being processed, the pulse will be cut off directly, the enabling will be turned off, and the brake will be turned on. This will cause impact to the robot and is not recommended.

2. If you need to pause (stop) the program while it is running, press the <||> key to slow down the system to stop the program running and the robot action. When the program is stopped in this way, all internal state, output port, counter, variable, etc. related to the program are maintained. On restart, press the <◎> key directly, and the program will continue to run normally.

3. In order to determine whether the program stops, press the <||> key and observe whether there is "Automatic stop" in the status bar .

8.2.3 Run in infinite loop

► The toggle control mode switch is in play mode (PLAY), as shown in Fig.8.9.



Fig.8.9

The status bar shows "Play ", and then changes the running mode 【Single Loop】 / 【Single Line Run】 to 【Infinite Loop】 , and changes the automatic speed to 【Automatic Speed 10%】 .

After the previous preparation is completed, press the <◎> key to run the program, and the system pops up a prompt box:

In the reproduction mode, press the Run button, you need to run the program, you need to press the Run button again, or click on the interface to give up.

Press < ◎ > key again to run, the running interface is shown in Fig.8.10.

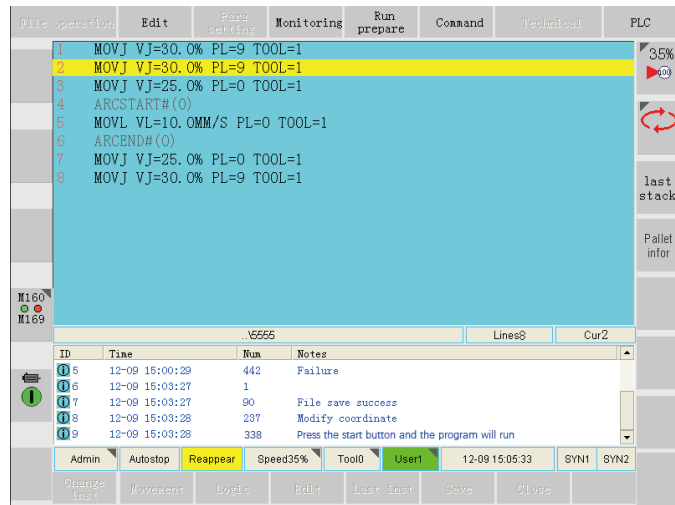


Fig.8.10

► Stop the robot during running:

If you need to pause (stop) during operation, press the < || > key to slow down the system and stop the program and the robot action.

► Adjust speed:

During operation, if you need to increase the automatic speed, you need to stop the robot first, and then operate the speed control button to achieve speed control function.

► Switch mode:

1. When the program is running, you can click the icon of running mode directly to switch back and forth between [Single Line Run] [Single Loop]. When you need to switch to Infinite Loop, you need to stop the program before switching.

2. In the pause (stop) state, click the running mode icon, click Select in the pop-up window, or press the physical buttons < + > and < - > on the right side of the mode icon to switch the running mode at will.

► When the program is running, the working mode is switched:

Currently in reproduction mode, if the program is running, you need to press the < || > key to stop the program running, and then switch the mode to the desired mode (instruction mode or remote mode).

► Stop program running in play mode

In this mode, the program stops running by pressing the stop key. System slowly stop program operation and robot action. When the program is stopped in this way, all internal state, output port, counter, variable, etc. related to the program are maintained. On restart, press the <◎> key directly, and the program will continue to run normally.

★Cation

1. Switch the mode switch to teaching mode or reproduction mode, and the program stops forcibly. When the system is being processed, the pulse will be cut off directly, the enabling will be turned off, and the brake will be turned on. This will cause impact to the robot and is not recommended.
2. If you need to pause (stop) the program while it is running, press the <||> key to slow down the system to stop the program and the robot action. When the program is stopped in this way, all internal state, output port, counter, variable, etc. related to the program are maintained. On restart, press the <◎> key directly, and the program will continue to run normally.
3. In order to determine whether the program stops, press the <||> key and observe whether there is "Automatic stop" in the status bar .

8.3 Emergency Stop

► During auto-running, if the robot is found to be working abnormally, the emergency stop button should be pressed quickly.

► After emergency stop, the current working status of the robot may be abnormal. When resetting the robot to alarm, special attention should be paid.

When the robot is in Play mode, and the program is running. After stopping the program with the emergency stop button, to start the robot again, follow these steps:

1. First, check whether the robot body, fixture, etc. are abnormal? Can you continue running the program?
2. Then rotate to release the emergency stop button; Press the 【R】 key to reset the current alarm information.
3. Click 【Serve down】 switch to 【Server on】 .
4. Reduce the speed of play mode and switch the running mode to run in a single line run mode.

5. Repeatedly press the< © >key to test if the program is working abnormally.
6. After confirming that the robot is working normally, increase the speed and switch the working mode to Infinite loop mode or single loop mode.
7. Click on the program key and the robot starts working.



WeChat public account



Tiktok



Reference material

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